

Payment.java - Explanation

ALGORITHM / EXECUTION FLOW:

1. When a new payment object is created (e.g. new CreditCardPayment), this class's constructor runs first.
2. It automatically assigns a unique ID and records the exact time.
3. It forces child classes to implement a specific method called process().

SECTION 1: SHARED STATE (Encapsulation)

'protected' means these fields are hidden from the outside world, but visible to child classes (like CreditCardPayment). This prevents accidental modification of transaction data.

SECTION 2: THE CONSTRUCTOR

This runs every time ANY payment is created. It automatically generates metadata for the transaction.

SECTION 3: THE POLYMORPHIC CONTRACT

'abstract' means there is no code block here. We are forcing every child class to write their own version of this method. When the PaymentProcessor calls payment.process(), the JVM looks at the actual object type at runtime and executes the correct child's version.